

DRY-RUN GUIDE | dojo_matrix.py

Source: Python\3rd_Belt for practice\Matrix\dojo_matrix.py

CORE BEHAVIOR

Checks that every element above the main diagonal is zero.

REPRESENTATIVE INPUT

[[1,0],[2,3]].

STEP TRACE

Only above-diagonal cell is [0][1]=0; all checks pass.

EXPECTED OUTPUT / RETURN

'Yes'

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\3rd_Belt for practice\Matrix\dojo_matrix.py; line numbers refer to the original file. No source was modified.

001 | # Question: Check whether a matrix is lower triangular.

002 | # Add your solution here.

003 |

004 | def is_lower_triangular_matrix(matrix):

005 | m = len(matrix)

006 | if m == 0:

007 | return True

008 | n = len(matrix[0])

009 |

010 | for i in range(m):

011 | for j in range(n):

012 | if i < j and matrix[i][j] != 0:

013 | return False

014 | return True

015 |

016 | try:

017 | m, n = map(int, input().split())

018 | matrix = [list(map(int, input().split())) for _ in range(m)]

019 |

020 | print("Yes" if is_lower_triangular_matrix(matrix) else "No")

021 |

022 | except ValueError:

023 | print("Invalid input. Please follow the input format: '<rows> <cols>' followed by the matrix rows.")

024 |

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.